



DEPARTMENT OF MINES
SOUTH AUSTRALIA
GEOLOGICAL SURVEY
EXPLORATION DIVISION

REPORT OF
EXPLORATORY GEOPHYSICAL SURVEYS AT MOUNT GUNSON -
PERNATTY LAGOON (S.M.L. 96)
- Austminex Pty. Ltd. -

by

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EXPLORATION GEOPHYSICS SECTION

29th September, 1966.

D.M. 1141/65

63/75

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SOUTH AUSTRALIA

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29th September, 1966.

Rept. Bk. No. 63/75
G.S. No. 3554
D.M. 1141/65

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ABSTRACT

Induced polarization, resistivity, gravity, magnetic, electro-magnetic, self potential and radio-active methods were used over areas in which information from drill logs was available at the Mount Gunson and Lagoon Workings to assess their suitability as exploration tools in search for new copper ore bodies or the delimiting of known deposits. Only the induced polarization and gravity methods yielded positive results, but these are not correlatable with the geology and pattern of mineralization known at present.

INTRODUCTION

Experimental geophysical surveys were requested by Austminex Pty. Ltd. to determine the suitability of the various available techniques as primary exploration tools for the location of new copper ore bodies and/ or the delimiting of known deposits on S.M.L. 96 in the Mount Gunson - Pernatty Lagoon district (Locality Plan 66-822).

During the period 2nd to 9th September 1966, a geophysical field party comprising B.E. Milton, Senior Geophysicist, J. J. Hussin, Assistant Senior Geophysicist, B. J. Taylor, Geophysical Assistant, and one field assistant carried out induced polarization, resistivity, gravity, magnetic, electromagnetic, self potential and radiometric measurements over surveyed lines of drill holes established by Austminex Pty. Ltd. Lines on which geophysical measurements were taken are shown on the Locality Plan (drawing 66-823).

PREVIOUS GEOPHYSICAL OPERATIONS

Aeromagnetic data has been published from a survey flown by the Bureau of Mineral Resources on behalf of the S.A. Department of Mines in 1962. Contour plans of total magnetic intensity for TORRENS (drawing 66-385) and Windabout (65-663/6) and profiles for TORRENS (66-720/6 and 721/6) are available.

During February and March 1964, 53½ miles of induced polarization traverses were laid down by Geoscience Inc., the leaseholder of S.M.L. 55. One line (30S) about 6 miles in length but not continuous profiling passes very close to line B of the Main Open Cut Workings, while the most northern line (33N) extends on to the Lagoon some distance south of the Lagoon Workings.

Electrode spacing used was 300 feet, and a number of generally weak anomalies were defined, the strongest of which is coincident with the strong anomaly on line B described in this report.

GEOLOGY

The geology of the Pernatty Lagoon - Mount Gunson District has been described by Dickinson (1942) and Johns (1964). The following details of the areas investigated during the surveys covered by this report are taken from Johns.

"Copper ore deposits comprise discontinuous, almost horizontal, irregular, tabular bodies confined to subhorizontal Sturtian Series sandstones, grits and dolomites. They are covered by up to 50 feet of overburden and appear to occupy two distinct levels, one approximately 70 feet above lagoon level and the other at that level.

The main orebody disclosed in the Mt. Gunson workings is 70 feet above lagoon level in sandstones and is up to 14 feet thick with grades up to 9 per cent Cu. The copper minerals include carbonates, chlorides, sulphides and silicate, occurring as a cement to well rounded grains of thin bedded friable sand-

stone. The ore horizon outcrops and has been exploited in Young's and the main open cut (100 feet by 60 feet by 7 feet) and it extends south-west to Brennan's workings where it closely follows a contact between red gritty sandstone and an overlying white sandstone member. To the east in Brennan's shaft, ore occurs in sandstone immediately above the dolomite.

The Gunyah Lake workings are situated $1\frac{1}{2}$ miles to the north of Mt. Gunson H.S. near the southern shore of an arm of the lagoon. Sandstones which outcrop on the floor of the lagoon, or have only a very thin cover, carry from 0.2 to 7.6 per cent copper as atacamite, covellite, bornite and chalcocite over a thickness of 5 feet. The ores occur as a cement to sand grains and in irregular masses of high grade sulphide from which 15 to 20 per cent copper ore was hand picked. Beneath the sandstone is a hard dense red quartzite which carries chalcocite in joint and bedding planes, but not universally impregnating the rock."

METHODS

Induced Polarization

The principle causes of induced polarization in rocks are the over-voltage phenomenon where metallic mineral grains are in contact with electrolytic solutions and electrolytic polarization which occurs in rocks having ion-exchange capacity due to distributed fixed charges. Induced polarization methods are useful in locating sulphides or other electronic conductors in rocks in which electrolytic polarization is small. The effect from oxidised copper minerals is not likely to be measureable.

The induced polarization contributed by overvoltage effects depends on the current density, the size and distribution of the sulphide particles, the resistivity of the matrix and country rock, and the total sulphide content. The specific surface

resistance of sulphides decreases rapidly as current density is increased. Disseminated particles of the order of 0.2 mm in diameter give the highest over-voltage effect. Sulphides occurring in nodules or in association with carbon may be ineffective in polarizing, and fairly large amounts (up to 10%) of sulphides are necessary to provide polarization in excess of the background level caused by electrolytic polarization. In a sulphide bearing rock, the amount of current which flows through the sulphide grains depends on the manner in which the grains are situated in the pores. With an oxide film, the grain may exhibit a large ohmic drop in potential at its surface.

If the resistivity of the matrix in which sulphide particles are distributed is very high, a decrease in matrix resistivity will increase the amount of polarization, but will decrease it if the matrix resistivity is initially very low. As a consequence, sulphides in sandstone present a more difficult problem in induced polarization prospecting than does sulphide mineralization in limestone or igneous rocks. In the latter type rocks, the high matrix resistivity assures current flow through the sulphide grains even when their surface resistances are high.

The relation of electrolytic polarization to clay content is only directly related when the clay is uniformly distributed throughout the pore structure of the matrix. Kaolinite will produce a maximum effect when it occupies from 5% to 10% of the pore spaces which are saturated with groundwater containing 100 ppm dissolved salts. Maximum polarization would be observed for large amounts of clay if a higher salinity groundwater were present. (Refer Anderson & Keller, 1964.)

Resistivity

The resistivity of rocks encompasses an extremely large range, being from 10^{-6} ohm-metres for minerals such as graphite to more than 10^6 ohm-metres for dry igneous rocks. Sandstone resistivities have been measured in a range varying from 30 to

more than 10^5 ohm-metres.

Most rocks and minerals are insulators in the dry state. In nature they almost always hold some intersitial water with dissolved salts and thereby acquire an ionic conductivity which depends on the moisture content, the nature of the electrolytes and the concentration thereof.

Some minerals such as graphite, pyrrhotite, pyrite, chalcopyrite, galena and magnetite are relatively good conductors. A dissemination of such minerals within a rock can also make it a better conductor, although a continuous ore body would produce a much higher conducting effect.

Gravity

Subsurface structures causing lateral irregularities in density distribution will give rise to corresponding variations, relative to a chosen datum, in the force of gravity. Structures having economic importance often alter the earth's normal gravitational field by less than one part in ten million, making necessary the use of highly refined and sensitive gravity meters, and accurate mapping and elevation control. From gravity data the location and magnitude of the corresponding density variations in the underlying rocks can often be estimated. Such estimates are never unique when obtained on gravity information alone, but on the basis of them areas can sometimes be singled out for further geophysical work or exploratory drilling.

Magnetic

Anomalous magnetic properties in subsurface rocks give rise to variations in the earth's magnetic field. The variations result from changes in depth of buried magnetic rock material, from changes in its susceptibility, or from both. Since few sedimentary rocks are appreciably magnetized, the magnetic method

of prospecting will generally give information only on igneous rocks or on ore deposits with magnetic constituents.

Electromagnetic

Electromagnetic waves attenuate in the earth at a rate depending on the frequency and the electrical characteristics of the earth, waves of higher frequency falling off in intensity more rapidly with depth. When they impinge on a conducting formation or ore body they induce currents in the conductors, in accordance with the laws of electromagnetic induction. These currents are the source of new waves which are radiated from the conductors and can be detected by suitable instruments on the surface. Any inhomogeneities in the electromagnetic field observed on the surface would indicate variations in the conductivity below and suggest the presence of anomalous masses in the subsurface.

Self Potential

The self potential method involves measurement on the surface of electric potentials developed in the earth by electrochemical action between metallic minerals and solutions with which they are in contact. No external electric fields are supplied. When different portions of an ore body are contiguous with solutions of different composition, chemical reactions take place which result in different solution pressures at the respective areas of contact. The difference in solution pressure gives rise to a potential difference which causes current flow in the ground.

For this phenomenon to be measurable, the ore body should intersect the water table. Usually disseminations carrying less than five percent total conductive sulphides are unlikely to yield interpretable results, although continuity, not volume, contributes most to the strength of the potentials generated.

(Note: "Solution pressure" is a measure of the tendency of a substance to go into solution).

Radioactivity

Since small concentrations of radioactive minerals are often associated with ore deposits of commercially valuable, non-radioactive substances, it is sometimes possible to use radioactivity to detect and map such deposits. Before such an approach can be of value, however, the fundamental principles concerning the concentration and localization of the radioactive minerals in and around the mineralized area must be determined.

Present detectors, geiger counters and scintillometers can detect very small quantities of gamma radiations. Gamma rays are emitted during radioactive disintegration processes, for part of the energy released during disintegration is converted into the additional mass that the emitted particle takes on when passing from its bound state in the nucleus to its free state, part of it is used to give the particle kinetic energy, and the rest is converted into gamma radiation.

Although the radioactive isotope of potassium constitutes only about 0.012 percent of all potassium, this element is very common in nature, and radioactive effects of potassium are often registered in prospecting operations.

RESULTS

Induced Polarization

Three lines in the Mount Gunson Main Workings area and two in the Lagoon Workings area were surveyed using the Induced Polarization technique. All work was carried out in the frequency domain with Geoscience instruments with a fifty feet dipole-dipole electrode configuration at frequencies of 3 c/s and 0.3 c/s. Some readings were also recorded at 0.1 c/s on the Lagoon Workings, but no significant difference to 0.3 c/s was noted. Traverses were situated along or near surveyed lines laid down by the lease

holder so that drill log information could be related to the I.P. results. Drill line B intersects open cut workings in the Mount Gunson Main Workings area and the I.P. traverse was offset by 50 feet to the west to avoid the workings and old and recent dump material.

Drawings 66-764, 765 and 766 display contours of Apparent Resistivity, Frequency Effect and Metal Factor on lines B, C and D in the Main Workings area, together with cross sections obtained from the drill logs. Drawing 66-763 shows contours of I.P. data only for two lines on the Lagoon Workings, one intersecting line G and the other along line F.

The results from the Main Workings area show a broad frequency effect anomaly on each line, with the effect arising from shallower sources and becoming much stronger to the east, i.e. increasing from line D to C to B. On line B the frequency effect anomaly lies between 100S and 300N, with the related metal factor contours showing a strong anomaly centred below 200N and suggestive of a shallow origin and a small, weaker anomaly at greater depth below 50S. It can be seen that there appears to be no consistent relation between the total copper assay recorded on the drilling logs and the I.P. effects on this line, or on lines C and D. Assays do not record the presence or otherwise of copper sulphides or of other possible sources, e.g. pyrite, of anomalous I.P. results, and as most of the recorded copper minerals in this area are non-sulphides, a direct relationship with recorded mineralization would not be expected. However, there could be a significant effect from "damp clay" intersected between 20 and 40 feet in hole B11 (a) (225N) and from "sandy clay" between 15 and 25 feet in B3. Both these holes are in close proximity to the metal factor anomalies referred to above.

The contribution to the I.P. effect of electrolytic polarization ("membrane effect") resulting from the presence of clay material has been discussed under "Methods". With the present paucity of geological data, interpretation of I.P. anomalous

results must remain tentative, but it appears possible that a relationship exists between the anomalies recorded on line B and electrolytic polarization. Even if the data are the resultant of two separate effects, one arising from sulphide mineralization and the other from the membrane effect, it would not be possible to discriminate between them. Similar results are apparent on lines C and D with lower frequency effect values and it is possible that the anomalous patterns on these traverses arise in whole or in part from electrolytic polarization. Deepening of the hole at B11 (a) to determine whether sulphide mineralization at depth contributes to the I.P. effect would enable more certain interpretation of these anomalies.

Evidence from the three traverses described suggests that anomalous I.P. results could be obtained over much of the Mount Gunson Main Workings area, but that such results probably could not be interpreted to indicate the presence or otherwise of copper mineralization, or indeed of sulphide mineralization, and that it would not be practical to use the technique as an exploration method in this environment with existing knowledge. It is suggested that any geological data obtained in this vicinity in the future be related to the I.P. patterns to determine if more certain correlation can be made.

At the Lagoon Workings, the first line was laid out partly on sandstone outcrop at the edge of the lagoon and partly on the lagoon surface in an effort to determine the value of the apparent resistivity under these varying conditions. As would be expected values are low and decrease away from the shore. Frequency effect and metal factor contours show very little evidence of anomalous effects.

A second line was centered on drill hole F3, where the highest value of mineralization was recorded in this area. (25 feet at 3.12%). Resistivity contours show extremely low values near surface with an increase in depth, which suggests an anisotropic effect due to the highly conductive salt layers at the

surface. Two frequency effect anomalies are revealed, one between 50S and 100S and the other between 50N and 200N. The former is closest to hole F2 in which no copper mineralization was recorded, while the latter lies between F3 and F5 in which relatively high values were noted. Again, the interpretation of these results will depend on available geological data; at present, the only obvious fact is the lack of correlation between the I.P. pattern and recorded copper mineralization.

Gravity

Mount Gunson Main Workings

Gravity stations were established at all identifiable drill holes shown on Austminex Pty. Ltd. drawing GP5 in the Main Open Cut Workings area using a World Wide gravity meter No. 101. These stations have been transferred to drawing 66-825, "Contours of Bouguer Gravity".

Observed gravity values were converted using a factor of .1060 milligals/ scale division. Drill hole levels, referred to an arbitrary datum, have been stated to be accurate to 0.1 feet and elevation corrections to obtain Bouguer anomaly values were applied at the rate of .061 milligals/foot (assuming a density of 2.59 gms/cc). Latitude corrections were calculated using a value of .0220 milligals/100 feet and the coordinates of the station (corrected for magnetic declination) shown on the Austminex plan. No topographic correction was considered necessary. Because of the very small residual gravity contrasts, instrumental drift was observed at intervals of no greater than one hour at two or more consecutive stations.

The Bouguer anomaly contour plan (drawing 66-825) shows a strong regional gradient falling off in a direction about 30° north of west, with no pattern which could be related to known mineralization or geology as recorded on drillers' logs. A resid-

ual gravity plan (drawing 66-826) was therefore constructed by removing a linear gradient of .034 milligals/ 100 feet decreasing in a direction $28\frac{1}{2}^{\circ}$ north of west to ascertain if the residual pattern was correlatable with geological data.

The residual plan has been contoured at an interval of 0.1 milligals, and has a total contrast of only 0.3 milligals, suggesting that it is strongly influenced by small variations in thickness of overburden, by deviations from the assumed density value of 2.59 gm/cc, and by any errors in surveying. An example is the gravity 'low' centred about G4, which probably results from the presence of a sand ridge in that vicinity. It would appear that the results would not be of value as an exploration method in this area.

Lagoon Workings

The same method was applied at the Lagoon Workings, with contours of Bouguer anomaly presented in drawing 66-824. These results show even less variation from the regional gradient than those from the Main Workings, even though one line (E) was extended to the south and north in an effort to detect any change from the linear effect. As a result, no further work was done on the results.

Resistivity

Four Schlumberger electrical soundings made on Lines D and C showed that the white sandstone (see drawings 66-765 and 766 for drill logs) has a general resistivity of 60 - 80 ohm-meters. At the Mount Gunson Workings, this was underlain by a much lower resistivity material. However, due to the low and variable electrode contact resistance of the order of 8 ohms, the electrical sounding graphs proved inadequate as quantitative guides for interpretation. Results have not been included because

of the uncertainty introduced by the unusual contact resistance.

Magnetic

Ground magnetic results showed no anomalous zones in either area, only scatter values of ± 10 gammas.

Electromagnetic

The electromagnetic results showed no anomalous zones, only scatter values of ± 1 degree of field tilt.

Self Potential

Self Potential results also displayed no anomalous zones, only scatter values of ± 15 millivolts.

Radioactivity

Radioactive values showed only a random scatter on the surface at both workings and on drill cuttings remaining at the holes. A normal, minor anomaly about 3 or 4 times background was detected in the main open cut at the Mount Gunson Workings and was associated with a thin bed of clay. This was traced at a number of locations in the open cut and could possibly provide a stratigraphic marker in this general location.

CONCLUSIONS AND RECOMMENDATIONS

Induced polarization, resistivity, gravity, magnetic, electromagnetic, self potential and radiometric methods showed no diagnostic relationships to the ore occurrence at either the Mount Gunson Main Open Cut or the Lagoon Workings. An increase in geological information may enable interpretation of induced polarization

patterns, but it would not appear that any of the techniques would be useful as a first exploration method in this area.

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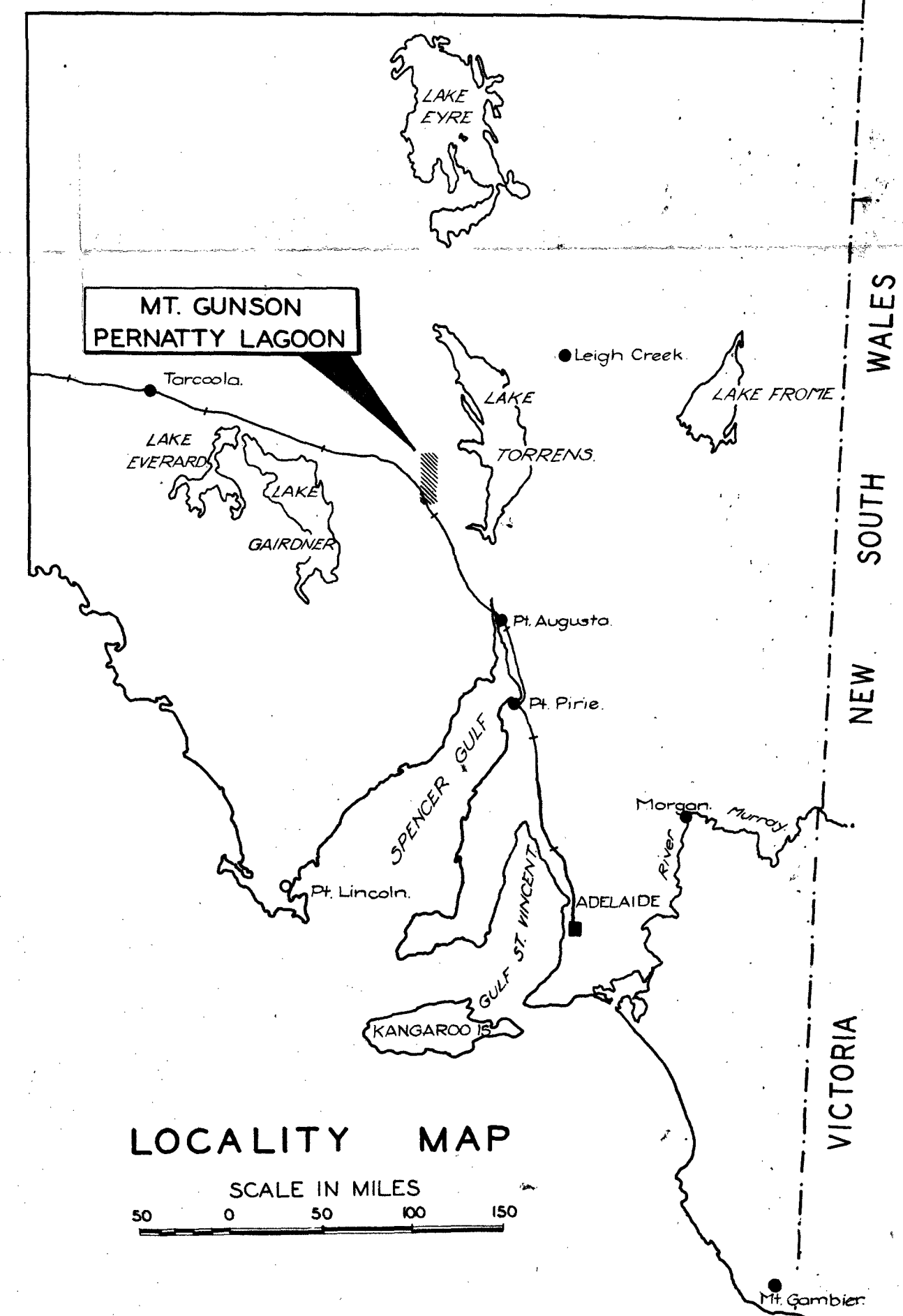
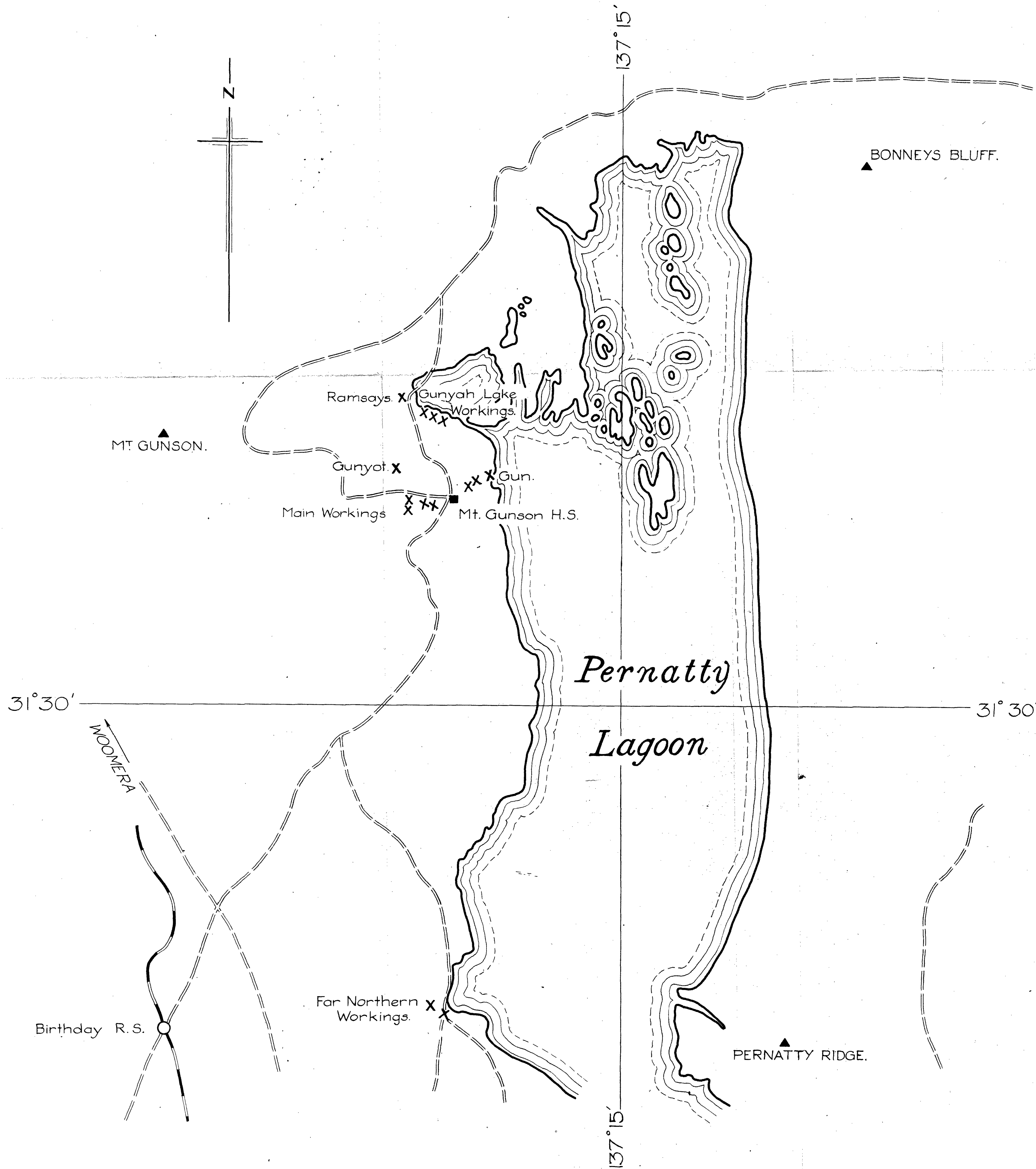
B. J. Taylor

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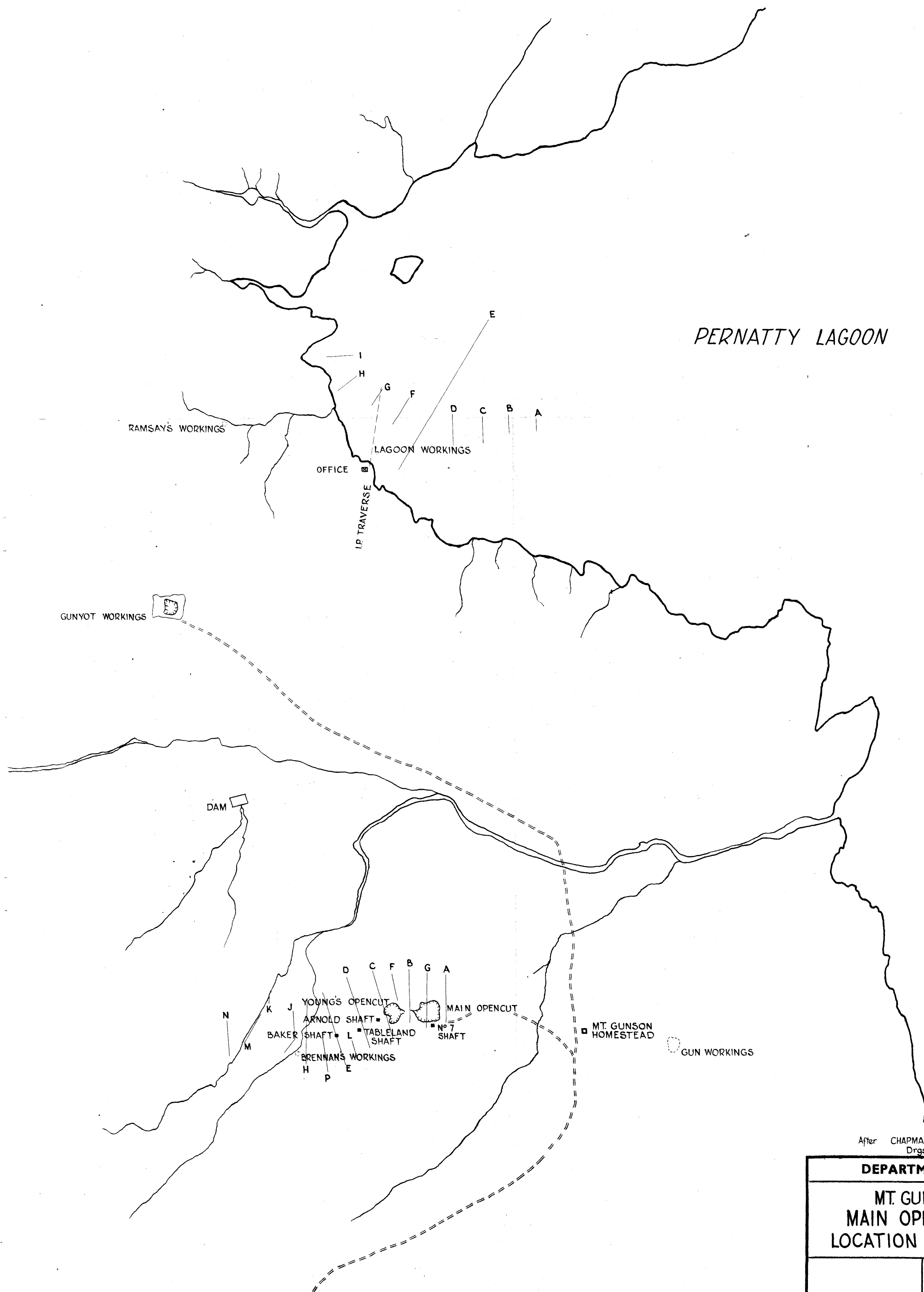
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29/9/1966

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DEPARTMENT OF MINES - SOUTH AUSTRALIA			
MT. GUNSON - PERNATTY LAGOON LAGOON WORKINGS LOCALITY PLAN.			
		Drn. B.M.	SCALE: 1" = 1 mile.
		Tcd. A.M.S.D.	66-822
		Ckd. L.V.W.	Ed.
Director of Mines		Exd.	DATE: 5-10-66.



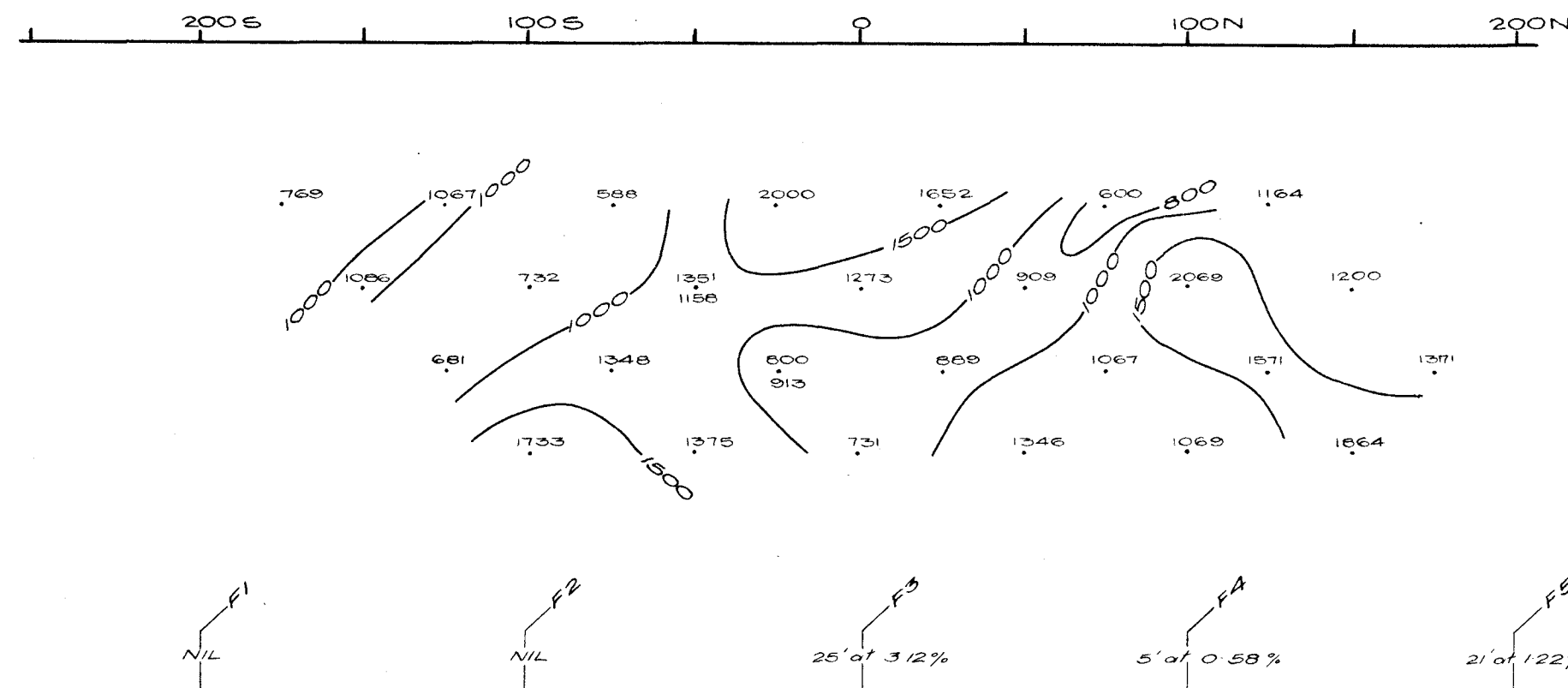
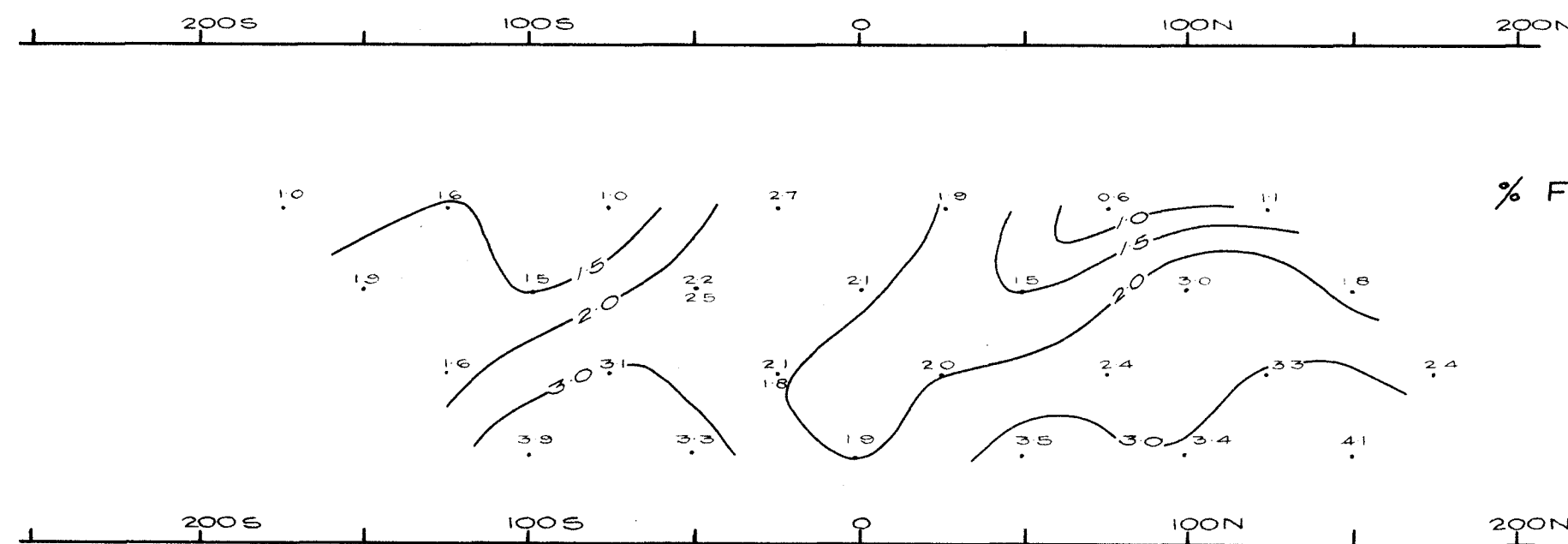
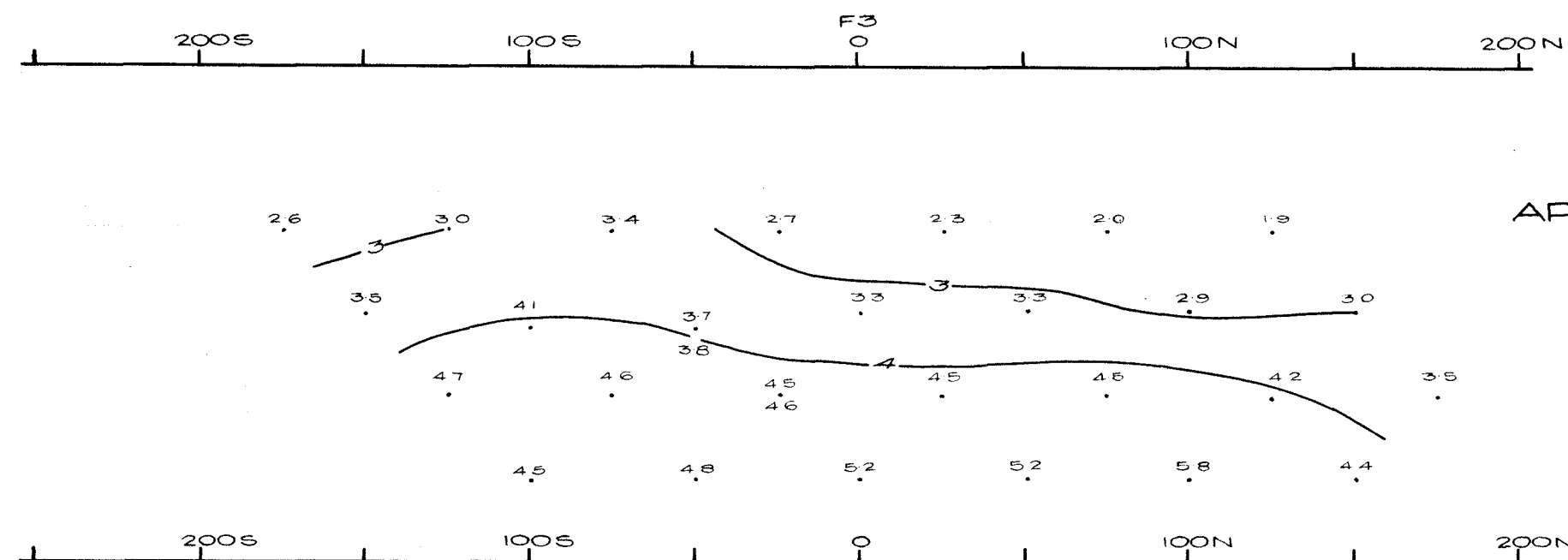
After CHAPMAN, WOODS, GRISWOLD & EVANS PTY. LTD.
Drgs. GP3, GP5, GP6

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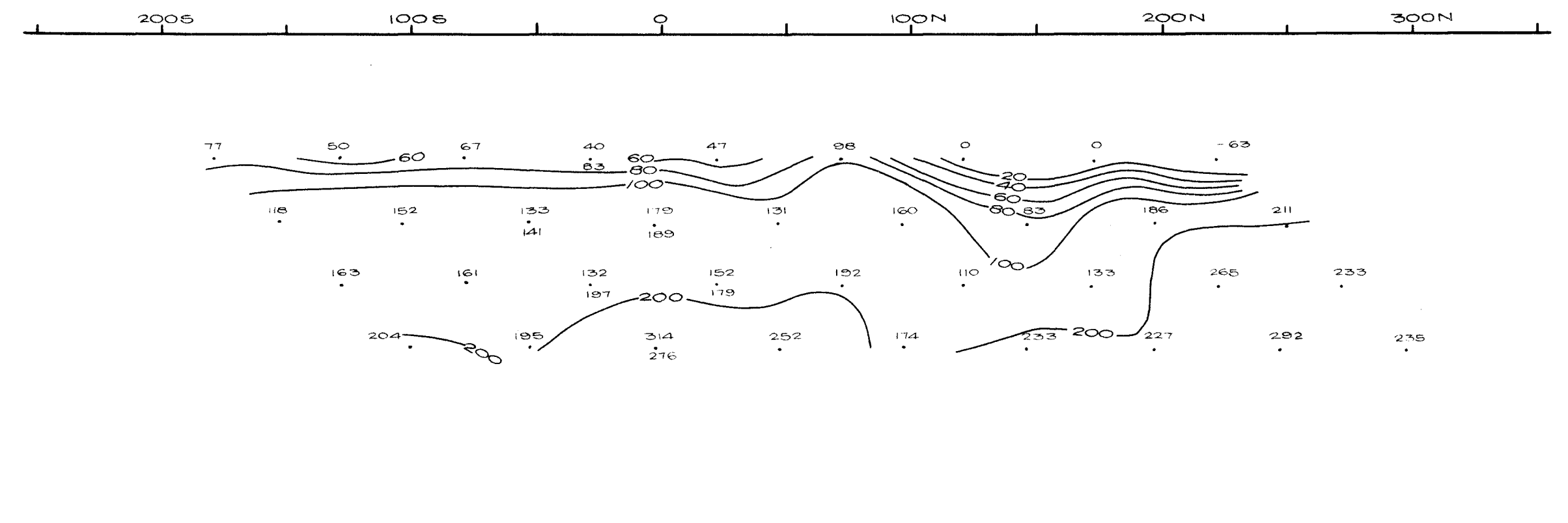
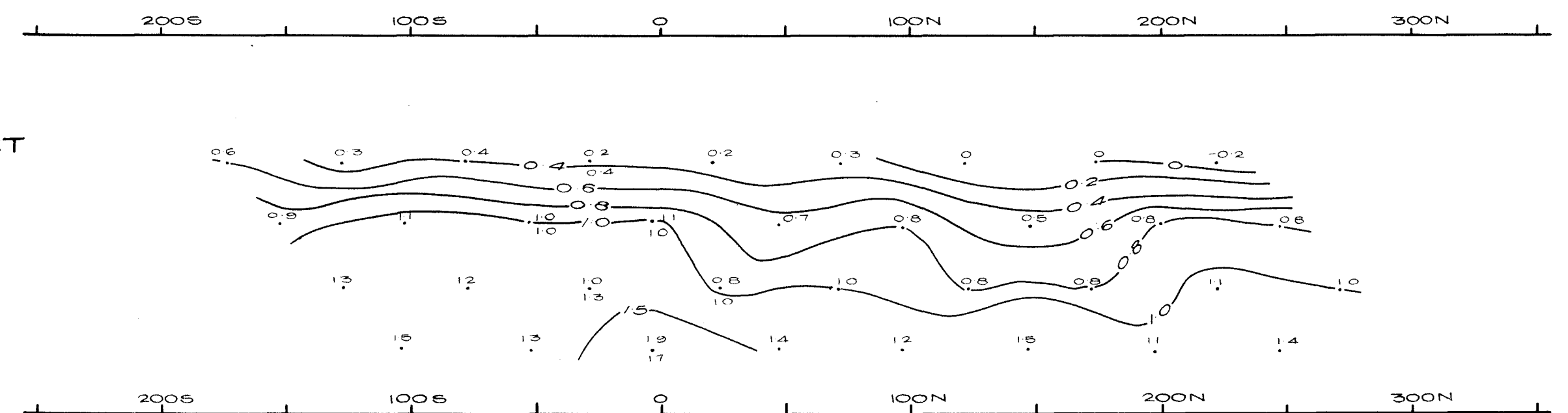
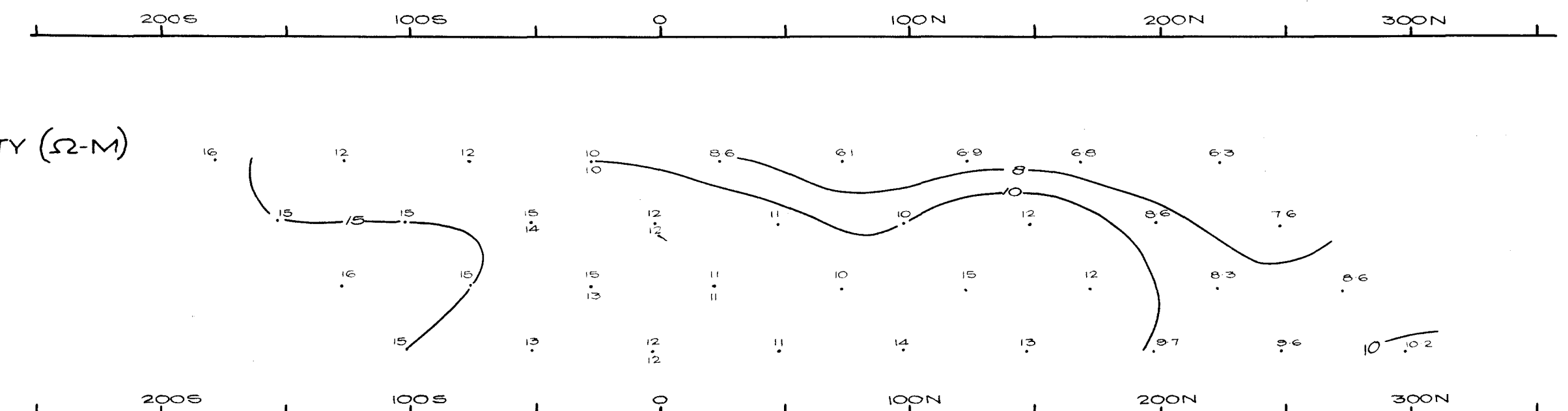
MT. GUNSON - PERNATTY LAGOON
MAIN OPEN CUT AND LAGOON WORKINGS
LOCATION OF GEOPHYSICAL TRAVERSES

Director of Mines		Drn.	SCALE: 1 INCH = 1000 FEET
		Tcd. ABH	66-823 Eb
		Ckd. L.V.W.	
		Exd.	DATE:

LINE F



LINE BETWEEN OFFICE AT 400S AND G₂ AT 460S



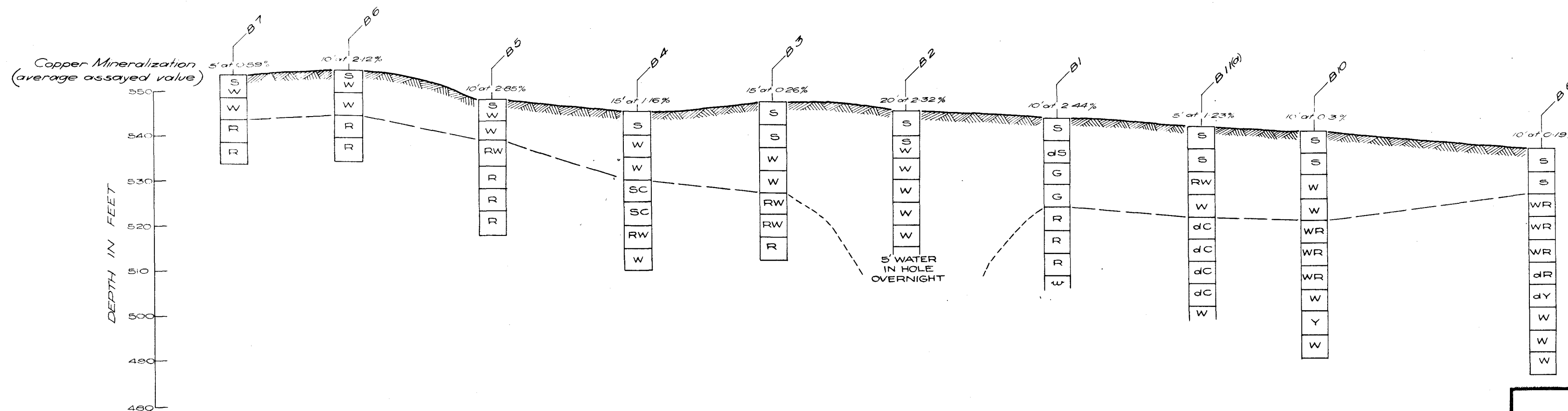
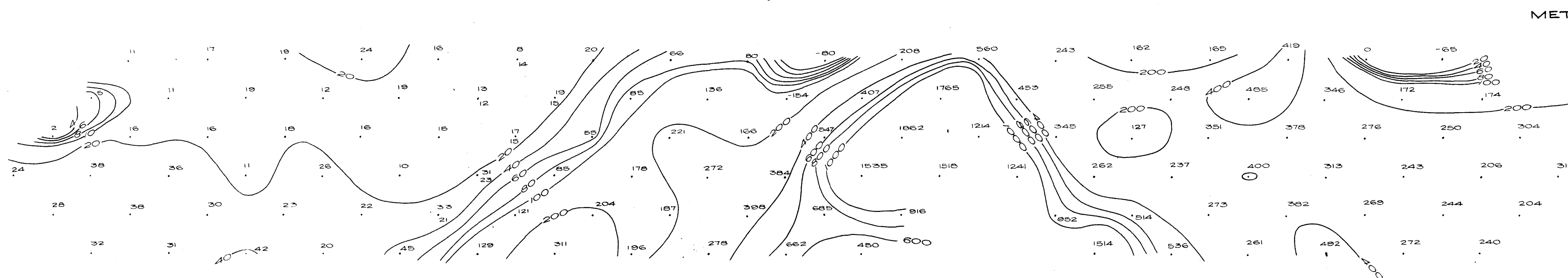
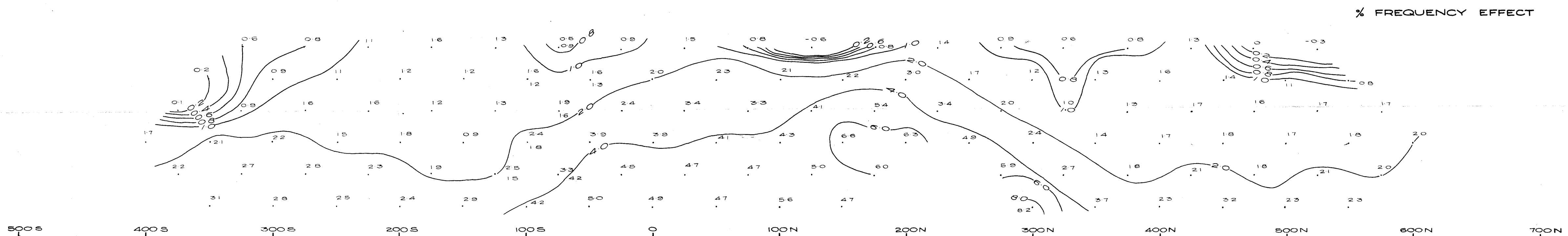
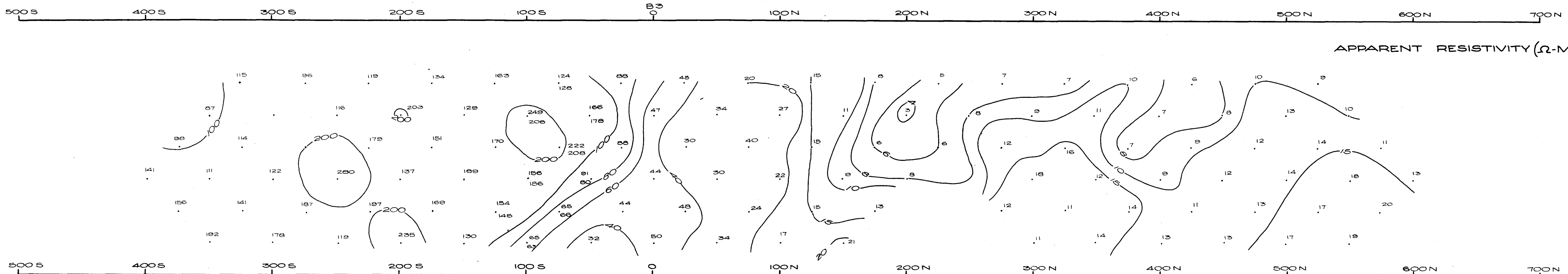
Copper Mineralization
(average assayed value)

DEPARTMENT OF MINES — SOUTH AUSTRALIA

INDUCED POLARIZATION RESULTS
GUNYAH LAKE WORKINGS
MT. GUNSON — PERNATTY LAGOON

Drn. B.M.	SCALE: 1 inch = 50 feet
Tcd. T.P.S.	66-763
Ckd. L.Y.W.	Ed
Exd.	DATE: 13-9-66

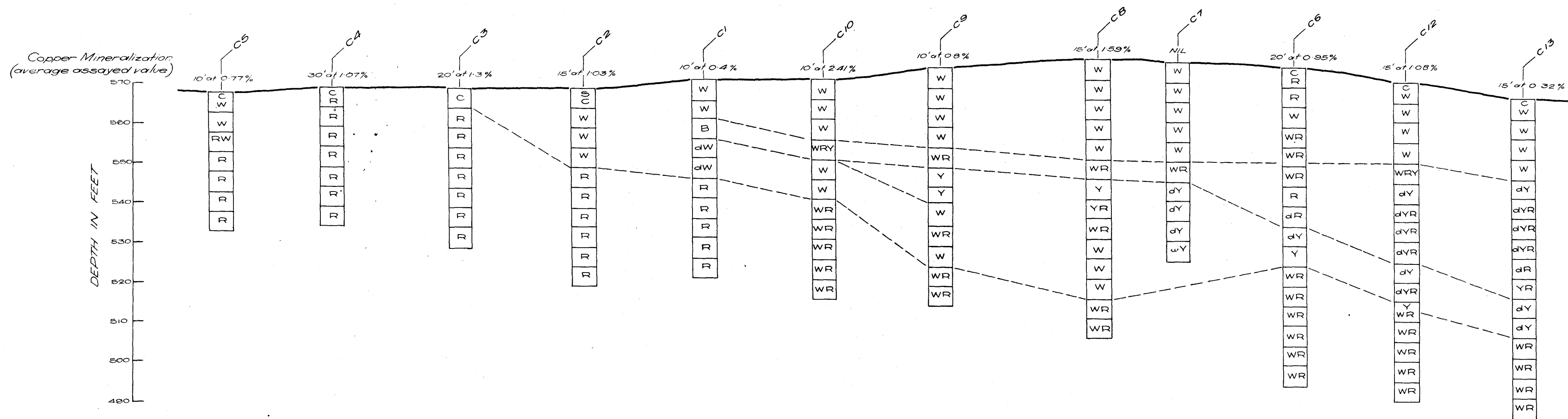
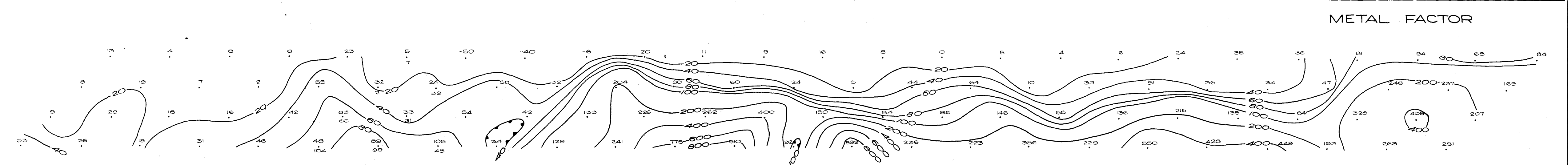
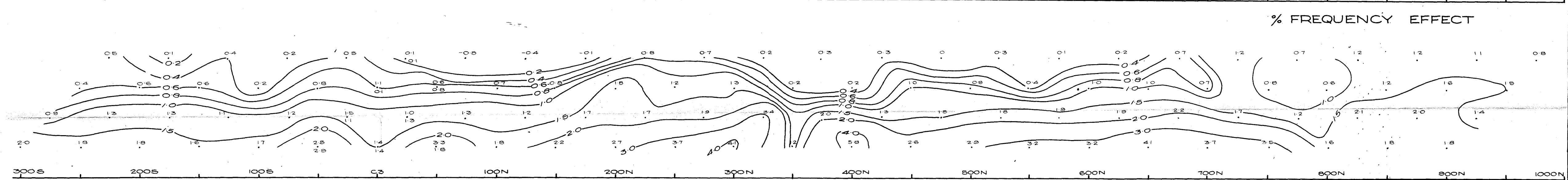
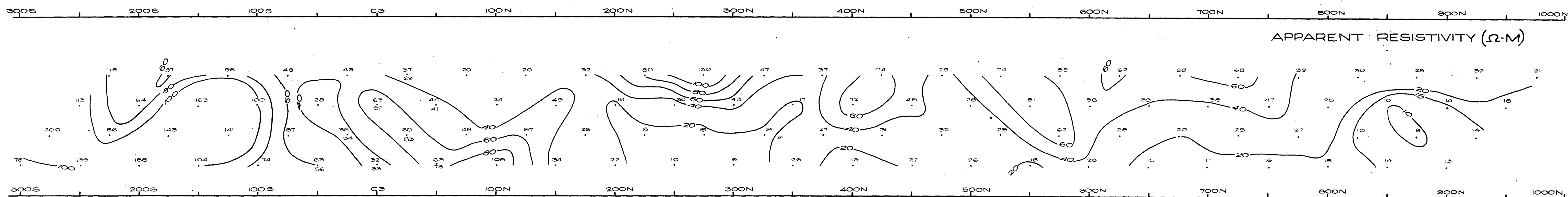
Director of Mines



LEGEND

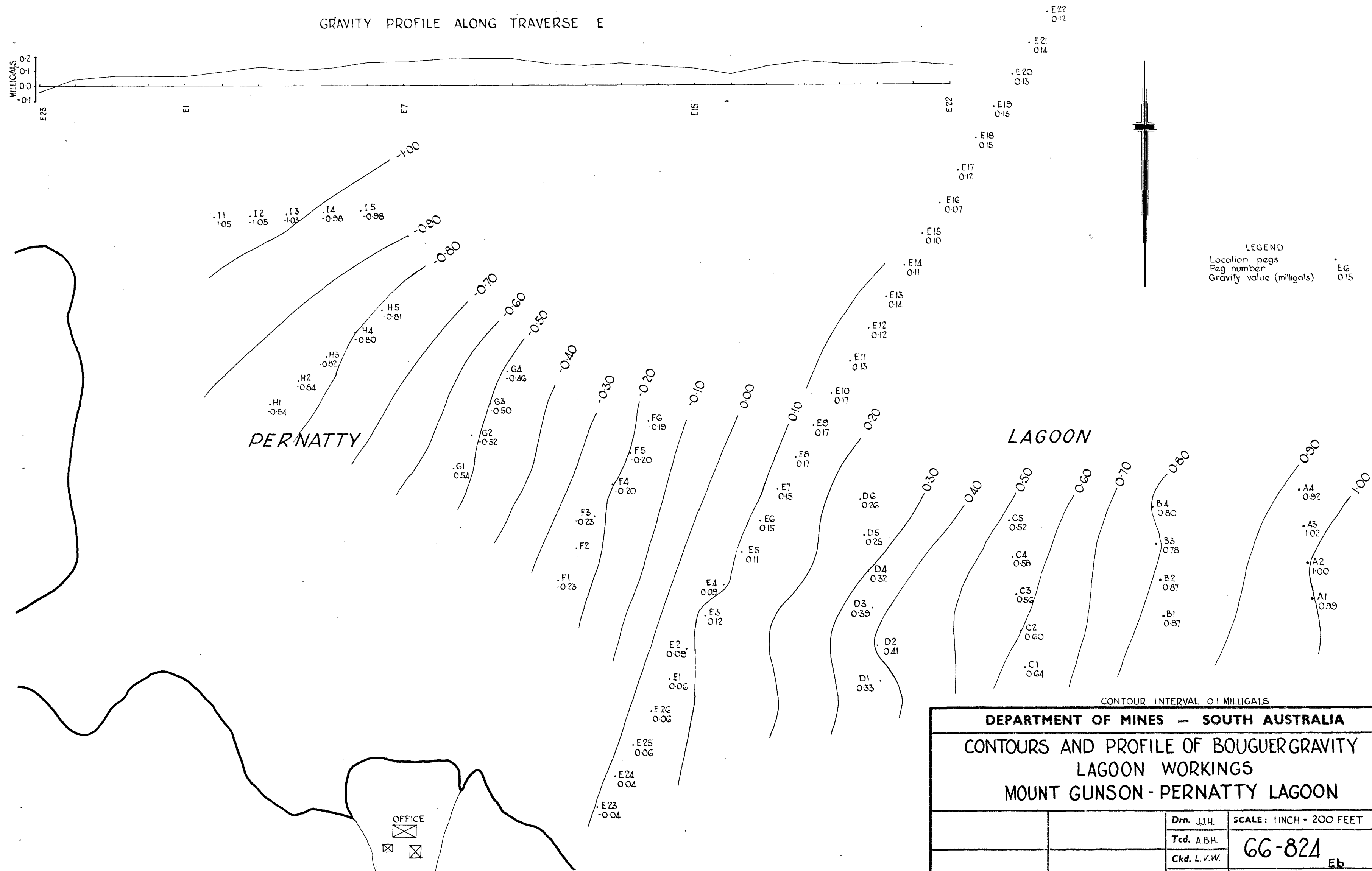
R = RED SANDSTONE
RW = RED-WHITE SANDSTONE
W = WHITE SANDSTONE
G = GREY SANDSTONE
Y = YELLOW SANDSTONE
C = CLAY
S = SAND
d = DAMP
w = WET

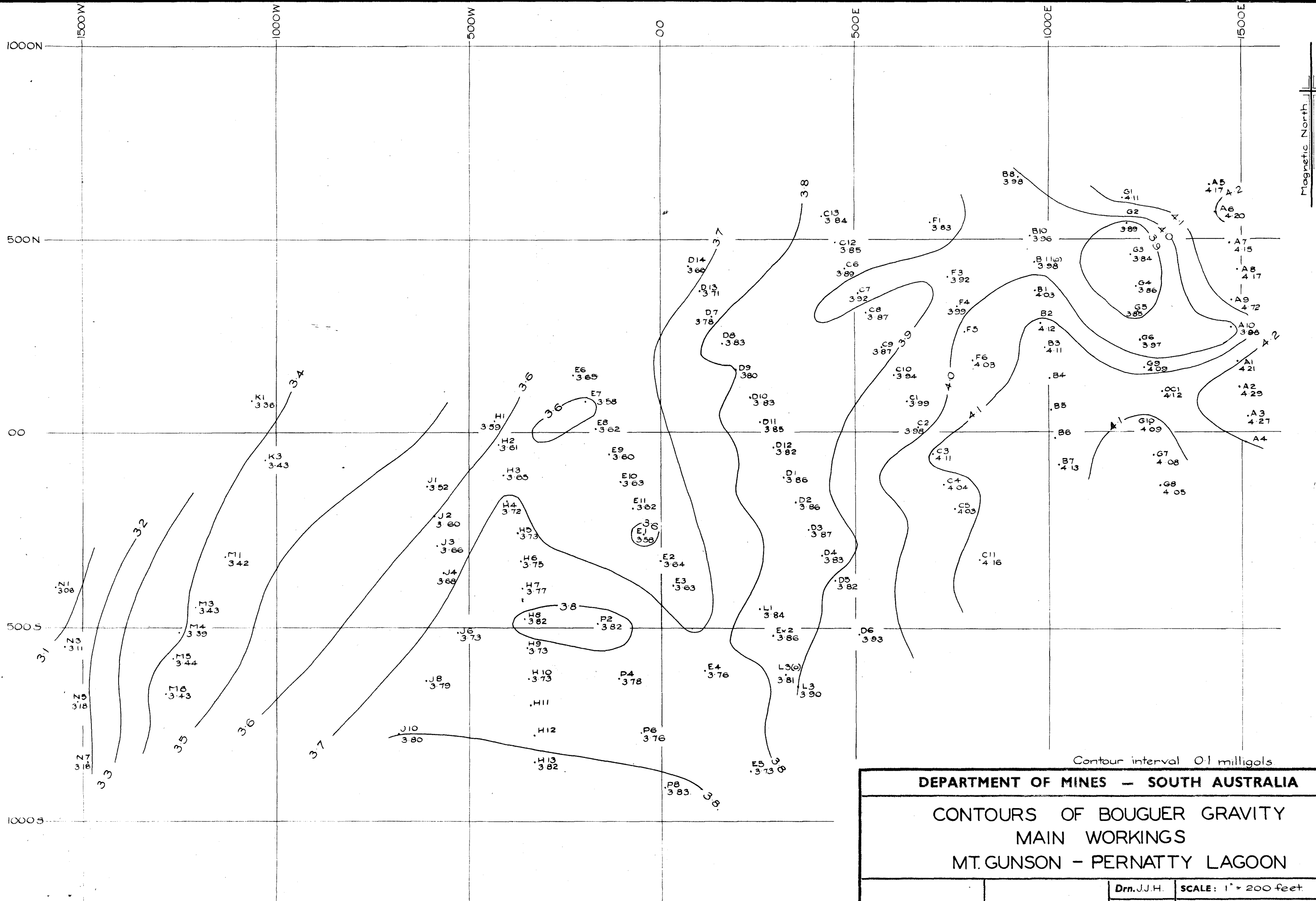
DEPARTMENT OF MINES — SOUTH AUSTRALIA			
INDUCED POLARIZATION RESULTS AND DRILL LOGS			
MT. GUNSON MAIN WORKINGS			
LINE B (OFFSET)			
Director of Mines	Drn. B M	SCALE: 1 inch = 50 feet	
	Tcd. T.P.S.	66-764	
	Ckd. L.V.W.	Eb	
	Exd.	DATE: 13-9-66	



DEPARTMENT OF MINES - SOUTH AUSTRALIA			
INDUCED POLARIZATION RESULTS AND DRILL LOGS			
MT. GUNSON MAIN WORKINGS			
LINE C			
Drn. B.M.	SCALE: 1 inch = 50 feet		
Tcd. TRS	66-765		
Ckd. L.V.W.	Eb		
Exd.	DATE: 13-9-66		
Director of Mines			

GRAVITY PROFILE ALONG TRAVERSE E





Contour interval 0.1 milligals.

DEPARTMENT OF MINES — SOUTH AUSTRALIA

CONTOURS OF BOUGUER GRAVITY
MAIN WORKINGS
MT. GUNSON — PERNATTY LAGOON

Director of Mines

Drn. J.J.H.

Tcd. AMSD

Ckd. L.V.W.

Exd.

SCALE: 1" = 200 feet

66-825
Eb

DATE: 7-10-66

